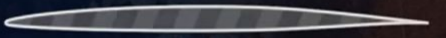
Airfoil of gliders

Airfoils

An airfoil is defined as the specific cross-sectional shape of a wing, blade, or sail designed to manipulate airflow and produce aerodynamic forces like lift and thrust



NACA 2412



NACA 66-206



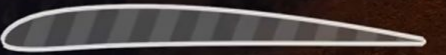
NACA 4421



NACA 0030



NACA 4412



ARA-D 10%



NACA 65(3)-618

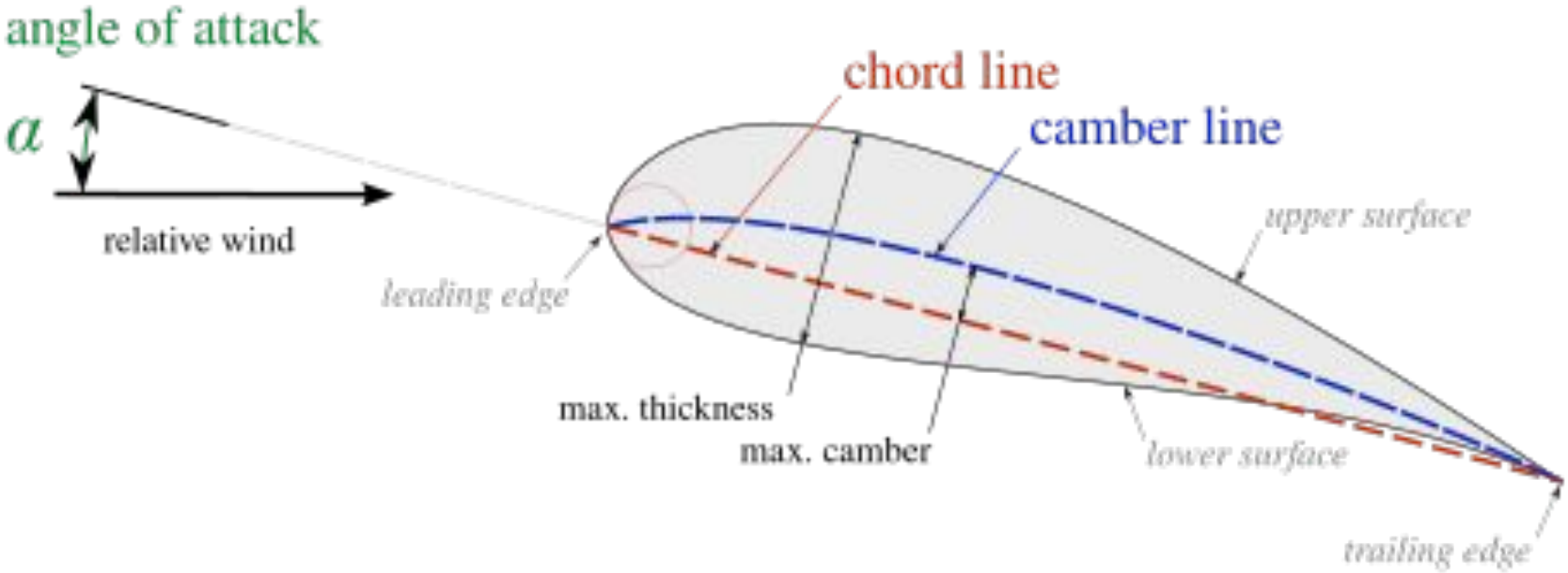


SC(2)-0712



DOUBLE WEDGE

THE EFFICIENT ENGINEER
SUBSCRIBE


angle of attack
α
relative wind
chord line
camber line
upper surface
leading edge
max. thickness
max. camber
lower surface
trailing edge

Angle of attack -
Angle of attack (AOA) is the angle between an airfoil's chord line (wing) and the oncoming relative wind

This parameter determines the amount of lift and drag
Increasing angle of attack means increasing lift and induced drag

Chord line -
The chord line is a straight reference line connecting the leading edge (front) to the trailing edge (back) of an airfoil section

Camber line -
The camber line (or mean camber line) is the curve halfway between an airfoil's upper and lower surfaces, running from the leading edge to the trailing edge

TYPES OF AIRFOILS

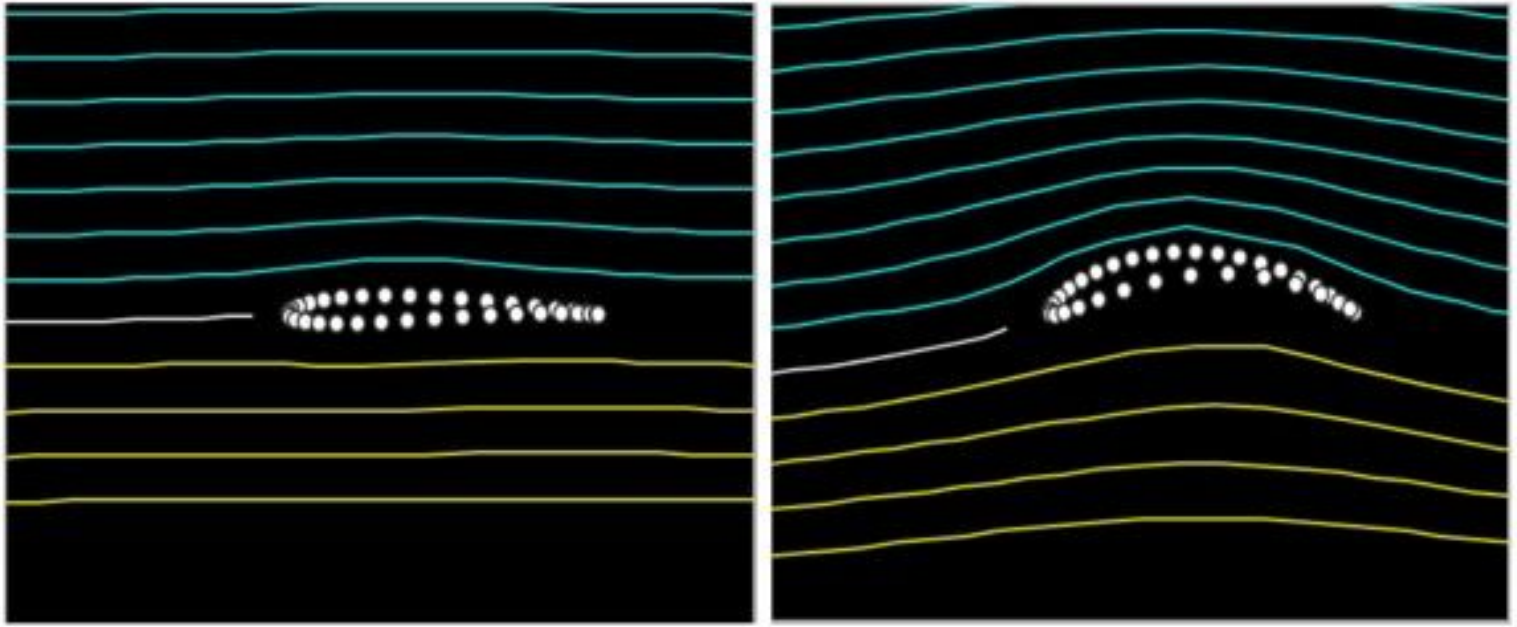
Symmetrical Airfoils:
-They have symmetric upper and lower surfaces
- same camber and chord line
- zero lift at zero angle of attack



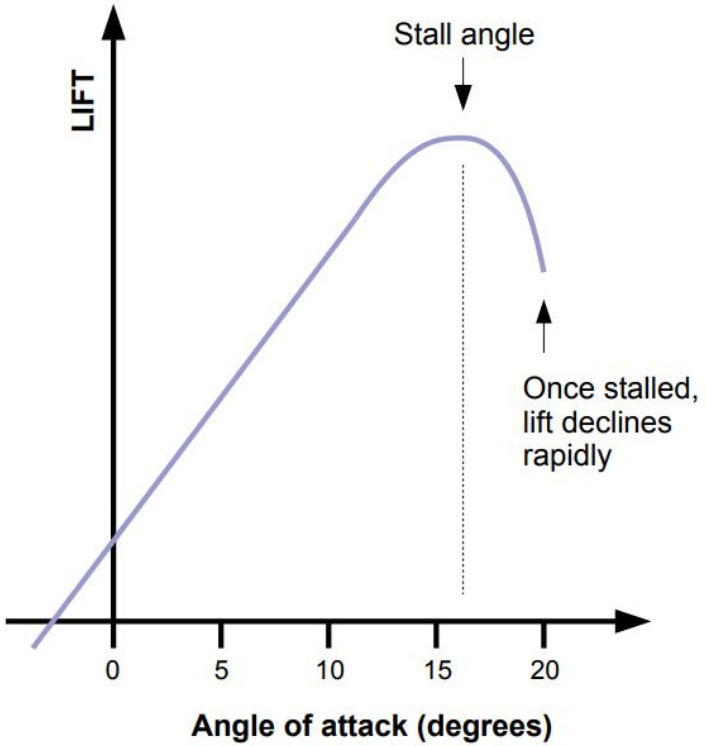
Unsymmetric / cambered (unequal curves)

The top curve is more pronounced than the bottom, allowing them to produce lift at zero angle of attack



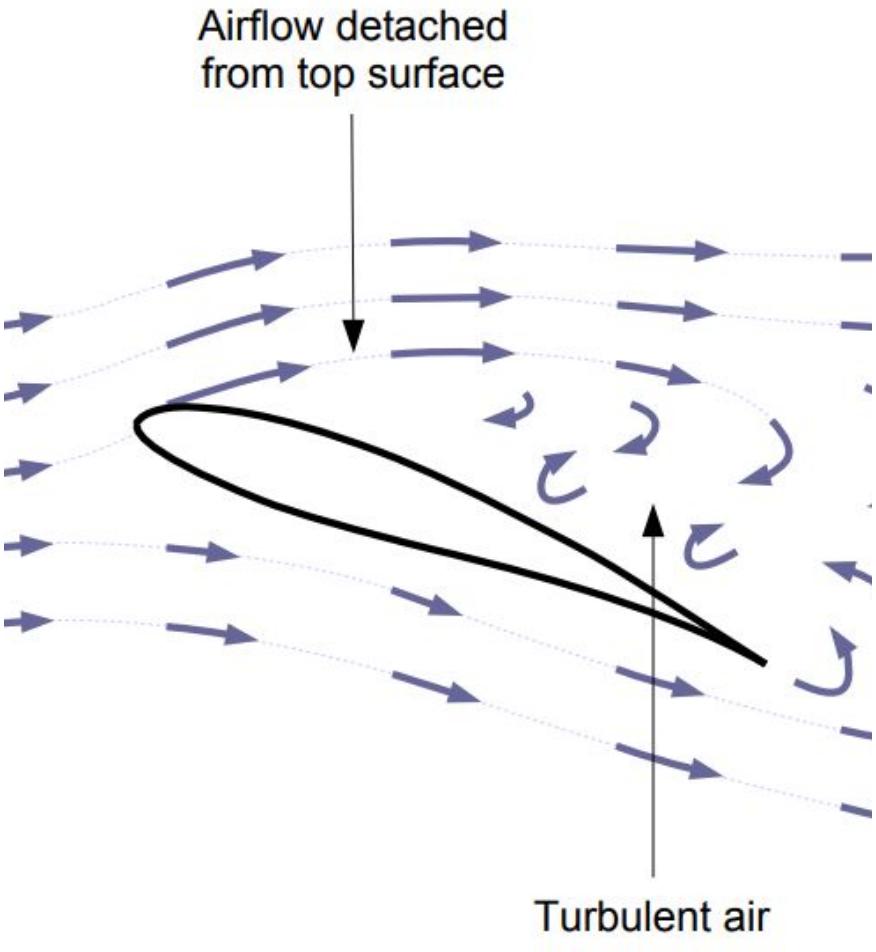


Flow turning at trailing edge is very important. More turning means more lift.



Lift generated increases linearly with the angle of attack

However increasing a wing's angle of attack also increases the amount of drag it produces
and eventually the wing will stall.



A wing **stall** occurs when the aircraft exceeds its critical angle of attack, causing a rapid separation of airflow over the wing's top surface, and air beneath the separated flow becomes turbulent.

This leads to a **significant loss of lift, a rise in drag, and often a sudden, involuntary nose-down pitch.**

NACA airfoils

NACA developed a series of thoroughly tested airfoils and devised four digit numbers that represented the airfoil section’s critical geometric properties.

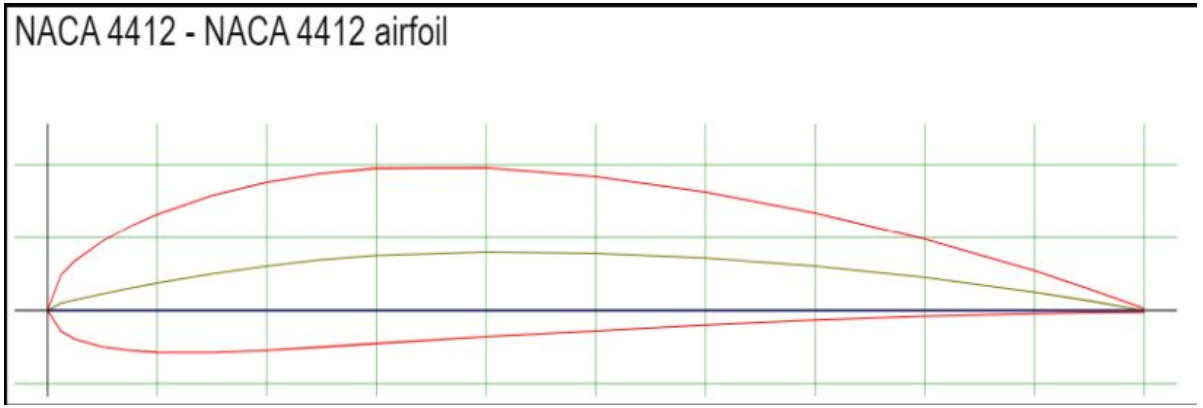
What does 4 numbers in NACA naming system mean?

First Digit: Maximum camber as a percentage of the chord length.

Second Digit: Location of maximum camber in tenths of the chord.

Last Two Digits: Maximum thickness as a percentage of the chord.

Example



NACA 4412

First Digit (4): Maximum camber is 4% of the chord length

Second Digit (4): Position of maximum camber is 40% of the chord from the leading edge.

Last Two Digits (12): Maximum thickness is 12% of the chord length

Types of gliders

Training Gliders

Used for beginner pilots

Aspect ratio 4.5-7.5

Stable and forgiving design

15-Metre Class Gliders
15 m wingspan
Flaps allowed
aspect ratios ranging from 21 to over 25

Standard Class Gliders

Maximum 15 m wingspan
high aspect ratios generally ranging from 20 to over 25

18-Metre Class Gliders
18 m wingspan
Very high glide ratio
Used in advanced competitions
aspect ratio between 27 and 33

Open Class Gliders
No wingspan limit
Extremely high performance
Can have very long wings
High Aspect Ratio: Ratios of 30–40+

Self-Launching (Motor Gliders)
Have small engines
Can take off without tow plane
Engine retracts after launch
high-aspect-ratio wings